

Lizzy Dale

+1(206)-890-0493 • lizzydale3@gmail.com • [LinkedIn](#) • [GitHub](#)

EDUCATION

University of Massachusetts Dartmouth | M.S. in Computer Science | *Sept 2025*

Coursework: Mobile Robotics, Computer Architecture, Algorithm Design, Machine Learning

University of Washington, Seattle | B.S. Oceanography | *June 2022*

SKILLS

- **Languages:** Python, Java, C/C++, Verilog, Bash, JavaScript
 - **Cloud & Backend:** AWS (S3, Lambda, Redshift), SQL, MongoDB, Node.js, Express, Flask, React, REST API Design, CI/CD, Linux
 - **AI & ML:** PyTorch, TensorFlow, Scikit-Learn, NLP, Deep Learning, Transfer Learning
 - **Hardware & Robotics:** Coppeliasim, Motion Planning, Embedded Systems
-

PROJECTS

Big Data ETL Pipeline | *AWS Glue, Redshift, Athena*

- Designed a serverless ETL pipeline for ingesting and transforming large sensor datasets, with AWS Lambda triggers and Glue jobs.
- Enabled schema evolution and optimized distributed queries for performance and scalability.

UK Web Change Tracker | *Flask, MongoDB, HTMX, Cloudflare Radar*

- Developed a full-stack platform to monitor web traffic shifts following the UK's 2025 age-verification law.
- Implemented RESTful APIs and an interactive HTMX/Tailwind dashboard with charts (time-series, bump, bar) and applied change-point detection and diff-in-diff analysis to measure shifts in domain rankings, traffic baselines, and VPN adoption

Autonomous Search & Rescue Robot Simulation | *Coppeliasim, Python, Motion Planning*

- Simulated a six-wheeled differential-drive robot navigating rubble using real-time sensor data, dynamic path planning, and heuristic route optimization
- Modeled real-time control loops and adaptive navigation behaviors, bridging concepts from robot kinematics and embedded systems.

Custom Processor Simulation | *Verilog, Computer Architecture*

- Designed a 5-stage pipelined CPU with out-of-order execution, speculative branching, and Tomasulo's algorithm.
 - Evaluated performance trade-offs in register renaming and branch prediction under variable load conditions.
-

EXPERIENCE

Woods Hole Oceanographic Institute | Engineering Assistant II | Jan 2023-Sept 2023

- Collaborated with teams on NASA's Sub-Mesoscale Ocean Dynamics Experiment mission on ocean dynamics to support field operations and mission planning
- Managed and organized 10+ years of legacy sensor data, troubleshooting broken pipelines and writing Python/Bash scripts to automate cleaning and validation
- Operated autonomous marine vehicles and oceanographic instruments (gliders, buoys, ship-based systems) for real-time data collection and integrated edge data into research workflows.
- Deployed & tested research instruments on *R.V. Sally Ride & Oscar Elton Sette* in multidisciplinary team setting